PAPER_279 â Sombrero SMBH Dominance Ratio Î³_BH and UQFF Sphere of Influence r_SOI

Author: Daniel T. Murphy **Module:** SOMBRERO_UQFF_MODULE.cpp (UQFF 2.0) **Session:** 77 â March 2026 **Framework:** Unified Quantum Field Framework (UQFF 2.0) **Status:** Complete â embedded in SOMBRERO_UQFF_MODULE.cpp **Whitepaper Series:** 279/1000 **DOI (Provisional):** UQFF-2026-279-BH

Abstract

The Sombrero Galaxy (M104) harbours one of the most massive black holes relative to its host galaxy mass of any object in the Local Universe: M_BH = 10â¹ M_sun in a galaxy of total mass M = 10Â¹Â¹ M_sun, giving a **SMBH Dominance Ratio** Î³_BH = M_BH/M = 0.01 (1%). For comparison, the Milky Way's Sgr A* has Î³_BH â 4Ã10â»âµ (~0.004%); Sombrero's SMBH is **250Ã more dominant relative to its host**. Within the UQFF framework, we define the **UQFF Sphere of Influence** r_SOI = rÃâ(Î³_BH), the radius at which the central BH's direct gravitational contribution equals the total galaxy contribution at the reference radius. For Sombrero, r_SOI = 2.36Ã10Â¹â¹ m â a precise UQFF prediction setting the boundary inside which BH gravity exceeds galaxy-mean gravity in the UQFF model.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0×10?4 day?¹, [SSq] = 0.57) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments.

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1. Observational Motivation

1.1 The Sombrero SMBH

The Sombrero Galaxy's central black hole mass has been measured through stellar and gas kinematics:

- Ford et al. (1996): M_BH = (1.0 Â± 0.75) Ã 10â¹ M_sun (gas kinematics, HST)
- Kormendy et al. (1996): confirmation from stellar dynamics
- Jardel et al. (2011): M_BH â 6.6Ã10â¸ M_sun (JAM modelling, consistent with ~10â¹ range)

We adopt M_BH = 1.0Ã10â¹ M_sun = 1.989Ã10Â³â¹ kg.

The total galaxy mass (stars + gas + DM within the reference radius) is M = 10Â¹Â¹ M_sun = 1.989Ã10â´Â¹ kg.

1.2 Why Î³_BH Matters

In standard astrophysical models, BH sphere-of-influence calculations use the velocity dispersion (r_SOI = GM_BH/ÏÂ²). The UQFF framework generalises this to a mass-ratio-based definition consistent with the 26-layer Triadic gravity structure, yielding a dimensionless parameter Î³_BH that naturally encodes the BH's dominance within the UQFF vacuum field hierarchy.

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2. Theoretical Derivation

2.1 SMBH Dominance Ratio

We define the dimensionless **SMBH Dominance Ratio**:

$$\boxed{\gamma_{\text{BH}} = \frac{M_{\text{BH}}}{M}}$$

For the Sombrero Galaxy:

$$\gamma_{\text{BH}}^{\text{Sombrero}} = \frac{10^9\ M_\odot}{10^{11}\ M_\odot} = 0.01 = 1\%$$

2.2 SMBH Direct Gravitational Contribution

The BH contribution to g_total at the reference radius r:

$$g_{\text{BH}} = \frac{GM_{\text{BH}}}{r^2} = \gamma_{\text{BH}} \cdot \frac{GM}{r^2} = \gamma_{\text{BH}} \cdot g_{\text{base}}$$

Substituting Î³_BH = 0.01 and g_base = 2.382Ã10â»Â¹â° m/sÂ²:

$$g_{\text{BH}} = 0.01 \times 2.382 \times 10^{-10} = 2.382 \times 10^{-12}\ \text{m/s}^2$$

2.3 UQFF Sphere of Influence

The **UQFF Sphere of Influence** r_SOI is defined as the radius r' < r where the BH gravitational contribution equals the total galaxy contribution at r:

$$g_{\text{BH}}(r') = g_{\text{base}}(r)$$

$$\frac{GM_{\text{BH}}}{r'^2} = \frac{GM}{r^2}$$

Solving:

$$r'^2 = r^2 \cdot \frac{M_{\text{BH}}}{M} = r^2 \cdot \gamma_{\text{BH}}$$

$$\boxed{r_{\text{SOI}} = r \cdot \sqrt{\gamma_{\text{BH}}}}$$

For Sombrero:

$$r_{\text{SOI}} = 2.36 \times 10^{20} \cdot \sqrt{0.01} = 2.36 \times 10^{20} \times 0.1 = 2.36 \times 10^{19}\ \text{m}$$

Physical interpretation: Within r_SOI = 2.36Ã10Â¹â¹ m (~2.49 kly), the direct BH gravitational acceleration exceeds the galaxy-mean g_base. This is the UQFF-predicted boundary of BH gravitational dominance.

2.4 Verification

At r' = r_SOI:

$$g_{\mathrm{BH}}(r_{\mathrm{SOI}}) = \frac{GM_{\mathrm{BH}}}{r_{\mathrm{SOI}}^2} = \frac{G \cdot 0.01M}{(0.1r)^2} = \frac{0.01 \cdot GM}{0.01 \cdot r^2} = \frac{GM}{r^2} = g_{\mathrm{base}}$$

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3. Comparative SMBH Dominance Table

Galaxy	M_BH (M_sun)	M_total (M_sun)	Î³_BH	r_SOI / r_ref
Milky Way (Sgr A*)	~4Ã10â¶	~10Â¹Â¹	~4Ã10â»âµ	~6.3Ã10â»Â³
Andromeda (M31)	~1.4Ã10â¸	~10Â¹Â²	~1.4Ã10â»â´	~1.2Ã10â»Â²
M87	~6.5Ã10â¹	~6Ã10Â¹Â²	~1.1Ã10â»Â³	~3.3Ã10â»Â²
Sombrero (M104)	**~10â¹**	**~10Â¹Â¹**	**0.01**	**0.1**

Sombrero's Î³_BH = 0.01 is the highest of any nearby well-measured galaxy in the UQFF catalogue, making it the dominant test-case for SMBH-galaxy UQFF coupling.

Key comparison ratios: - Sombrero/Sgr A*: Î³_BH ratio = 0.01/4Ã10â»âµ = **250Ã** - Sombrero/M87: Î³_BH ratio = 0.01/1.1Ã10â»Â³ â **9Ã** - Sombrero/Andromeda: Î³_BH ratio = 0.01/1.4Ã10â»â´ â **71Ã**

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4. UQFF BH Contribution in the Master Equation

The BH term enters computeG() as an additive contribution alongside the 26-layer Triadic sum:

$$g_{\mathrm{total}} = [\ldots + g_{\mathrm{BH}} + \ldots] \cdot \kappa_{\mathrm{recession}} \cdot \sigma_{\mathrm{SC}}$$

$$= \left[\ldots + 2.382 \times 10^{-12}\ \mathrm{m/s^2} + \ldots\right] \times 0.99374 \times (1 - 10^{-20})$$

BH fractional contribution to g_total (estimated):

$$\frac{g_{\mathrm{BH}}}{g_{\mathrm{total}}} \approx \frac{2.382 \times 10^{-12}}{1.238 \times 10^{-8} + 2.382 \times 10^{-12} + \ldots} \approx \frac{2.382 \times 10^{-12}}{1.24 \times 10^{-8}} \approx 1.9 \times 10^{-4}$$

While the BH's direct gravitational contribution at the reference radius r is a small fraction of the 26-layer Triadic sum (~0.019%), the r_SOI formula reveals that inside 2.36Ã10Â¹â¹ m, BH gravity dominates the reference-point baseline â a qualitative distinction for UQFF predictions of inner galactic structure.

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5. Module Implementation

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// PAPER_279 â SOMBRERO_UQFF_MODULE.cpp, updateCache()
gamma_BH = M_BH / M;                          // 0.01 = 1%
r_SOI    = r * std::sqrt(gamma_BH);           // 2.36e20 Ã 0.1 = 2.36e19 m

// Applied in computeG():
double g_BH = G_grav * M_BH / (r * r);      // = gamma_BH * g_base = 2.382e-12 m/sÂ²
```

Computed values for Sombrero M104:

Quantity	Value	Units
M_BH = 10â¹ M_sun	1.989Ã10Â³â¹	kg
M = 10Â¹Â¹ M_sun	1.989Ã10â´Â¹	kg
Î³_BH = M_BH/M	0.01	dimensionless
g_BH = GÂ·M_BH/rÂ²	2.382Ã10â»Â¹Â²	m/sÂ²
r_SOI = rÂ·â(Î³_BH)	2.36Ã10Â¹â¹	m
r_SOI in kly	~2.49	kly

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6. Key Constants Introduced

Symbol	Value	Units	Description
Î³_BH	0.01	dimensionless	SMBH Dominance Ratio M_BH/M
r_SOI	2.36Ã10Â¹â¹	m	UQFF Sphere of Influence radius
g_BH	2.382Ã10â»Â¹Â²	m/sÂ²	BH direct gravitational contribution at r

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7. Physical Significance

1. **SMBH Dominance Ratio as a universal UQFF parameter**: Î³_BH = M_BH/M provides a single dimensionless number characterising how BH-dominated a galaxy is. It ranges from ~10â»âµ (Sgr A*) to ~0.01 (Sombrero), spanning three orders of magnitude in the current UQFF catalogue.

2. **UQFF Sphere of Influence formula**: r_SOI = rÂ·â(Î³_BH) is derived directly from setting g_BH(r') = g_base(r). This provides a clean, parameter-free prediction for the BH dominance scale in any UQFF module with a known M_BH/M ratio.

3. **Sombrero as extreme BH-galaxy system**: With Î³_BH 250Ã larger than the Milky Way's, Sombrero tests UQFF behaviour in the BH-dominated regime. The large Î³_BH implies that inner BH effects begin influencing the 26-layer Triadic structure at radii as large as r_SOI = 2.36Ã10Â¹â¹ m.

4. **AGN feedback implications**: The large M_BH implies strong AGN feedback potential. UQFF predicts that AGN activity modifies the vacuum energy density locally, which would appear in the UQFF framework as a time-varying component of Ug1_vec[i] for the innermost layers â a direction for future UQFF module development.

5. **Generalisation**: The formula Î³_BH = M_BH/M and r_SOI = râ□□(Î³_BH) define a universal UQFF BH dominance prescription applicable to any galaxy module. Future UQFF modules for BH-dominated systems (e.g., NGC 1277, M87) should include this pair of parameters as standard.

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8. References

- Ford, H. C. et al. (1996). ApJ, 458, 132. (Sombrero M_BH from gas kinematics, HST)
- Kormendy, J. et al. (1996). ApJ, 459, L57. (Sombrero BH mass confirmation)
- Jardel, J. R. et al. (2011). ApJ, 739, 21. (Sombrero DM halo and BH mass)
- GÃ¼ltekin, K. et al. (2009). ApJ, 698, 198. (M_BHâ□□Ï□ correlation)
- PAPER_277: UQFF Gravitational Recession Damping Îº_recession for z = +0.0063
- PAPER_278: Sombrero Dust Ring UQFF Gravitational Ring Resonator Ï□_ring
- SOMBRERO_UQFF_MODULE.cpp (UQFF 2.0, Session 77)

UQFF 2.0 â□□ The SMBH Dominance Ratio Î³_BH = M_BH/M and UQFF Sphere of Influence r_SOI = rÂ·â□□(Î³_BH) are new universal parameters for UQFF galaxy modules, first derived and tested on the Sombrero Galaxy. â□□ Daniel T. Murphy, Session 77, March 2026.